

WORKSHEET - NAPLAN Year 9

Number and Algebra

Fractions, Decimals and Percentages (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. ☒ Number and Algebra, NAP-95619

Cameron paid \$1.44 for 9 pencils.

How much did each pencil cost?

\$0.12 \$0.16 \$1.35 \$1.53

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2. ☒ Number and Algebra, NAP-95618

A swim squad normally do 60 laps per training session.

On a light training day, they only do two thirds of the laps of a normal session.

How many laps do they do in a light session?

20 35 40 45 50

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3. ☒ Number and Algebra, NAP-89734

David and Helena had identical ice-blocks.

David ate $\frac{5}{6}$ of his ice-block.

Helena ate more of her ice-block than David.

What fraction could Helena have eaten?

$\frac{3}{4}$ $\frac{6}{7}$ $\frac{7}{9}$ $\frac{8}{10}$

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4. ☒ Number and Algebra, NAP-89738

Which one of these is equal to $2\frac{4}{9}$?

$6 \div 9$ $8 \div 9$ $18 \div 9$ $22 \div 9$ $36 \div 9$

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5. ☒ Number and Algebra, NAP-42728

A tennis racquet costs \$95.

A fee of 10% is added to the price.

Which calculation will give the new price?

$95 + 10$ $95 + 0.1$ 95×0.1 95×1.1

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6. ☒ Numbers and Algebra, NAP-36811

The table below shows all the people at Angus' birthday party.

	Male	Female
Adult	15	25
Child	40	20

What fraction of the children at the party are female?

$\frac{20}{45}$ $\frac{20}{60}$ $\frac{45}{60}$ $\frac{20}{100}$ $\frac{45}{100}$

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7. ☒ Number and Algebra, NAP-42722

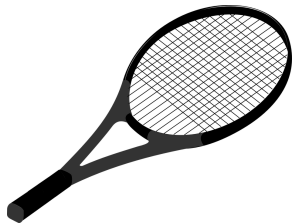
Simplify: $\frac{2 \times 4^3 \times 5^3}{4 \times 5^2 \times 4^2 \times 2^2}$

10 $\frac{5}{2}$ 1 $\frac{1}{4}$

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8. ✕ Number and Algebra, NAP-30897

Angus bought a tennis racket which cost \$22.50 before the Goods and Services Tax (GST) was added.



How much did Angus' tennis racket cost after adding the **10%** GST?

\$

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Worked Solutions

1. Number and Algebra, NAP-95619

$$\begin{aligned}\text{Cost of 1 pencil} &= \frac{1.44}{9} \\ &= \$0.16\end{aligned}$$

2. Number and Algebra, NAP-95618

$$\frac{2}{3} \times 60 = 40$$

3. Number and Algebra, NAP-89734

$$\frac{6}{7} \text{ is the only fraction greater than } \frac{5}{6}.$$

4. Number and Algebra, NAP-89738

$$22 \div 9$$

5. Number and Algebra, NAP-42728

$$\begin{aligned}\text{New price} &= 95 + (10\% \times 95) \\ &= 95 + 0.1 \times 95 \\ &= 95 \times 1.1\end{aligned}$$

6. Numbers and Algebra, NAP-36811

$$\begin{aligned}\text{Fraction of the children that are female} &= \frac{\text{Female Children}}{\text{Total Children}} \\ &= \frac{20}{60}\end{aligned}$$

7. Number and Algebra, NAP-42722

$$\frac{5}{2}$$

8. Number and Algebra, NAP-30897

$$\begin{aligned}\text{Cost} &= 22.50 + 10\% \times 22.50 \\ &= 22.50 + 2.25 \\ &= \$24.75\end{aligned}$$